

Artificial Intelligence and Machine Learning

Artificial intelligence (AI) is intelligence demonstrated by machines, as opposed to the natural intelligence displayed by animals including humans. AI research has been defined as the field of study of intelligent agents, which refers to any system that perceives its environment and takes actions that maximize its chance of achieving its goals.

Machine learning (ML) is a field of inquiry devoted to understanding and building methods that 'learn', that is, methods that leverage data to improve performance on some set of tasks. It is seen as a part of artificial intelligence. Machine learning algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to do so.

Retrieval-Augmented Generation (RAG)

Retrieval-augmented generation (RAG) is a technique for improving the quality of an LLM-generated response by grounding the model on external sources of knowledge. RAG models combine the strengths of pre-trained language models and information retrieval systems to produce more accurate and contextually relevant text.

By retrieving documents or snippets related to a user's prompt, RAG reduces the likelihood of hallucinations (generating false or incorrect information) and ensures that the model can reference specific, verified documents. This is especially useful for enterprise search, question answering over private documentation, and domain-specific assistants.